

Detecting Hidden Demand Shifts Before Established Categories Change

PROBLEM ID	ML-T2-011	DOMAIN	Retail Analytics · Unsupervised ML	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Retail systems record product purchases across stores, regions and time. Product demand is commonly interpreted through predefined categories, historical sales patterns and periodic reporting. These structures are useful, but they can conceal gradual changes in what customers are actually seeking.

02 THE RESEARCH PROBLEM

Investigate whether machine learning can discover structural changes in purchasing behaviour before those changes become obvious through conventional category-level reporting. The objective is to identify emerging behavioural regimes rather than simply forecast the next sales value.

03 WHY THE PROBLEM IS NON-TRIVIAL

A genuine demand shift can initially resemble ordinary noise. Seasonal effects, promotions, stock availability and price changes can create similar patterns. A method that detects every distribution change will produce many false signals.

CENTRAL RESEARCH QUESTION

“Can an ML system discover a market shift before predefined business categories recognize it?”

06 THE ENVISIONED OUTCOME

A software analytics system that surfaces candidate demand shifts, shows the products or behaviours contributing to them, and provides evidence about their persistence and uncertainty.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong solution will demonstrate that detected changes are more than isolated spikes and will critically examine whether the discovered structure is useful to a decision-maker.

04 THE RESEARCH OPPORTUNITY

Study clustering, representation learning, distribution-shift detection and temporal segmentation. Determine how products or customers should be represented, establish a historical baseline, and design a time-aware evaluation that distinguishes persistent shifts from temporary fluctuations.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- What constitutes a meaningful demand shift?
- How can legitimate seasonality be separated from structural change?
- Can representation learning reveal patterns hidden by predefined categories?
- How should a newly discovered segment be validated?
- How early can a shift be identified without using future information?

08 INVESTIGATION STANDARD

Problem understanding → Literature → Data discovery → Formulation → Baseline → Controlled experimentation → Evaluation → Iteration → Prototype → Technical paper → Defense

09 RESOURCE BOUNDARY

The core project must be completed at zero cost using free/open-source software and legally usable public or responsibly generated data. Paid APIs, paid datasets, proprietary software, dedicated hardware and paid cloud infrastructure are not required.

10 METHODOLOGICAL INDEPENDENCE

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RESEARCH & ENGINEERING OUTPUTS		
Literature review	Data & dataset documentation	Problem formulation
Baseline implementation	Experiment log	Evaluation & error analysis
Technical research paper	Working software demonstrator	Final presentation + defense

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Detecting Coordinated Information Operations Through Behavioural Structure

PROBLEM ID	ML-T2-012	DOMAIN	NLP · Graph ML · Social Computing	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Information can spread through large networks of accounts, pages and communities. Individual posts may appear ordinary while the timing, repetition and relationships among accounts reveal coordinated dissemination.

02 THE RESEARCH PROBLEM

Investigate whether machine learning can identify potentially coordinated information-spreading behaviour by combining content, temporal and relational signals. The project should focus on coordination patterns rather than attempting to determine the truth of every individual claim.

03 WHY THE PROBLEM IS NON-TRIVIAL

People naturally share the same news, repeat phrases and respond at similar times. Coordination can therefore be difficult to distinguish from genuine collective behaviour. Labels may also be incomplete and highly dependent on external investigations.

CENTRAL RESEARCH QUESTION

“Can coordination patterns reveal potentially misleading information operations even when individual posts appear legitimate?”

06 THE ENVISIONED OUTCOME

A software investigation interface that maps suspicious propagation patterns, highlights supporting evidence and ranks candidate coordinated activity for human examination.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong solution will avoid equating correlation with intent and will carefully discuss the limitations of inferring coordination from public behavioural data.

04 THE RESEARCH OPPORTUNITY

Review misinformation detection, graph anomaly detection, temporal coordination analysis and NLP representation methods. Establish a content-only or simple behavioural baseline, then investigate whether network structure adds meaningful evidence without introducing hindsight leakage.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- What should count as coordination?
- Which signals are strongest when content is individually plausible?
- How should organic collective behaviour be separated from suspicious synchronization?
- Can graph structure improve detection beyond text alone?
- How should uncertainty and human review be represented?

08 INVESTIGATION STANDARD

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Predicting Which Software Defects Are Likely to Become Critical

PROBLEM ID	ML-T2-013	DOMAIN	Software Engineering · Predictive ML	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Large software projects accumulate issue reports containing descriptions, components, timestamps, severity, developer activity and resolution history. Development teams must prioritize limited engineering attention across many unresolved defects.

02 THE RESEARCH PROBLEM

Investigate whether historical project data can be used to estimate which unresolved defects are likely to become high-impact or require significant intervention. The challenge is to predict future importance from information available before the outcome becomes obvious.

03 WHY THE PROBLEM IS NON-TRIVIAL

A defect’s eventual impact can depend on technical location, user exposure, release timing and development activity. Historical issue data can contain inconsistent labels and information that only became available after the defect was resolved.

CENTRAL RESEARCH QUESTION

“Can the future impact of a software defect be estimated from information available before its consequences become obvious?”

06 THE ENVISIONED OUTCOME

A software engineering analytics tool that ranks unresolved defects according to estimated future impact and provides evidence behind the ranking.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong project will show realistic temporal validation and explicitly identify features that would constitute leakage in a real development environment.

04 THE RESEARCH OPPORTUNITY

Study defect prediction, issue mining and software repository analytics. Establish a baseline using information available at issue creation, then investigate whether temporal, textual and project-level signals add predictive value. Carefully design experiments to avoid using post-outcome information.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- What should 'critical' mean in the selected project?
- Which issue information is legitimately available at prediction time?
- Does textual information add value beyond metadata?
- How should time-based validation be designed?
- Can the model provide useful prioritization rather than only a classification label?

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Following a Phishing Campaign as Its Messages Change

PROBLEM ID	ML-T2-014	DOMAIN	Cybersecurity · NLP · Clustering	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Phishing campaigns may generate many messages that differ in wording, domains, URLs and presentation while pursuing the same underlying objective. Treating each message independently can obscure relationships between events that belong to a common campaign.

02 THE RESEARCH PROBLEM

Investigate whether machine learning can group related suspicious messages and identify an evolving phishing campaign even when its individual messages continually change.

03 WHY THE PROBLEM IS NON-TRIVIAL

Attackers can modify wording and infrastructure rapidly. Legitimate organizations may also send many similar messages. Public datasets can contain duplicate campaigns or labels that do not fully describe campaign relationships.

CENTRAL RESEARCH QUESTION	“Can machine learning identify an evolving phishing campaign even when its individual messages continually change?”
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06 THE ENVISIONED OUTCOME

A software-based phishing campaign analysis system that groups related activity, displays campaign timelines and highlights features connecting apparently different messages.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong solution will show that campaign-level analysis adds information beyond individual-message classification.

04 THE RESEARCH OPPORTUNITY

Study text representation, clustering, similarity search, graph construction and temporal grouping. Establish an individual-message baseline, then investigate campaign-level representations. Design evaluation around grouping quality and temporal coherence rather than only binary classification accuracy.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- What evidence indicates that two messages belong to the same campaign?
- How can changing URLs or wording be handled?
- Which representations preserve campaign identity?
- How should clusters be evaluated when ground-truth campaign labels are incomplete?
- Can the system reveal campaign evolution over time?

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Estimating Road Risk Without Waiting for Accidents

PROBLEM ID	ML-T2-015	DOMAIN	Transportation · Geospatial ML	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Road safety analysis frequently relies on historical crash records. Yet severe crashes are relatively rare, and a low observed accident count does not necessarily imply that a road segment is inherently safe.

02 THE RESEARCH PROBLEM

Investigate whether publicly available geospatial, environmental and contextual data can be used to estimate latent road-risk patterns before accident history alone becomes statistically informative.

03 WHY THE PROBLEM IS NON-TRIVIAL

Risk is influenced by road geometry, traffic, speed, visibility, weather, intersections and temporal conditions. Accident records are also affected by reporting practices and exposure: a road with more traffic naturally has more opportunities for incidents.

CENTRAL RESEARCH QUESTION

“Can road risk be estimated from environmental and behavioural signals before accident history becomes statistically obvious?”

06 THE ENVISIONED OUTCOME

A software mapping and analytics prototype that produces interpretable road-risk estimates and allows users to inspect the evidence behind high-risk segments.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong solution will distinguish correlation from causation and avoid presenting a model-generated risk score as a definitive safety judgment.

04 THE RESEARCH OPPORTUNITY

Study spatial risk modelling, exposure-aware analysis and geospatial machine learning. Determine how risk should be defined and what public signals can support that definition. Compare a simple historical baseline with models that incorporate environmental or structural context.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- How should risk be normalized for traffic exposure?
- Which geospatial features contain meaningful information?
- How should spatial leakage be prevented?
- Can a low-accident road still be identified as potentially risky?
- How should uncertainty be represented on a risk map?

08 INVESTIGATION STANDARD

Problem understanding → Literature → Data discovery → Formulation → Baseline → Controlled experimentation → Evaluation → Iteration → Prototype → Technical paper → Defense

09 RESOURCE BOUNDARY

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Teaching a Machine Learning System When Not to Trust Its Own Prediction

PROBLEM ID	ML-T2-016	DOMAIN	ML Reliability · Uncertainty Estimation	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Machine learning systems are often evaluated by how accurately they predict on representative test data. In real deployment, however, inputs can differ from the conditions seen during training. A model may still produce a confident output even when its prediction is unreliable.

02 THE RESEARCH PROBLEM

Investigate methods for identifying situations in which an ML prediction should be considered uncertain, unfamiliar or unsuitable for automated decision-making.

03 WHY THE PROBLEM IS NON-TRIVIAL

Accuracy alone does not reveal whether a model knows when it is operating outside its reliable region. Confidence scores can be poorly calibrated, and different types of distribution shift can have very different consequences.

CENTRAL RESEARCH QUESTION

“Can an ML system recognize when it should not trust its own prediction?”

06 THE ENVISIONED OUTCOME

A software framework that produces predictions together with a meaningful reliability signal and demonstrates behaviour under both familiar and shifted conditions.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong solution will demonstrate that uncertainty estimates correspond to actual error behaviour rather than merely reporting a confidence number.

04 THE RESEARCH OPPORTUNITY

Study calibration, uncertainty estimation, out-of-distribution detection, abstention and reliability analysis. Select a suitable prediction task and establish a conventional baseline. Then construct controlled distribution shifts and investigate whether the system can identify when its own predictions should be treated cautiously.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- What does 'not trustworthy' mean for the selected task?
- How should confidence be evaluated rather than assumed?
- Which distribution shifts are detectable?
- When should a model abstain?
- Does uncertainty information improve decision quality?

08 INVESTIGATION STANDARD

Problem understanding → Literature → Data discovery → Formulation → Baseline → Controlled experimentation → Evaluation → Iteration → Prototype → Technical paper → Defense

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Reconstructing Missing Sensor Behaviour Without Inventing Reality

PROBLEM ID	ML-T2-017	DOMAIN	Time-Series ML - Data Reconstruction	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Real-world sensor systems frequently contain gaps caused by network interruptions, equipment failures, storage problems or data-processing errors. Missing values are often filled using simple interpolation or averages, but such methods can destroy important temporal relationships.

02 THE RESEARCH PROBLEM

Investigate whether machine learning can reconstruct missing portions of multivariate time-series data while preserving meaningful system behaviour.

03 WHY THE PROBLEM IS NON-TRIVIAL

The correct missing value may depend on relationships among multiple signals and on the system's temporal state. Long missing intervals are fundamentally more uncertain than isolated gaps. Evaluation is also difficult because the true missing value is unavailable in real deployment.

CENTRAL RESEARCH QUESTION

“Can an ML system reconstruct missing observations without inventing behaviour that was never present?”

06 THE ENVISIONED OUTCOME

A software reconstruction pipeline that fills controlled missing segments, quantifies reconstruction error and analyses how recovered data affects downstream modelling.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong project will avoid treating imputation accuracy alone as sufficient and will investigate whether reconstructed values preserve useful system behaviour.

04 THE RESEARCH OPPORTUNITY

Study imputation, sequence modelling, matrix completion and reconstruction methods. Create controlled missingness by hiding known observations, while distinguishing between missingness patterns that are random and those that resemble real sensor failures.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- Which missingness mechanisms should be simulated?
- How does reconstruction quality change with gap length?
- Do multiple correlated signals improve recovery?
- How should uncertainty in long-gap reconstruction be communicated?
- Does a numerically accurate reconstruction preserve downstream task performance?

08 INVESTIGATION STANDARD

Problem understanding → Literature → Data discovery → Formulation → Baseline → Controlled experimentation → Evaluation → Iteration → Prototype → Technical paper → Defense

09 RESOURCE BOUNDARY

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Understanding When Learning Engagement Begins to Change

PROBLEM ID	ML-T2-018	DOMAIN	Educational Technology · Sequential ML	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Digital learning environments record interactions with videos, documents, quizzes, coding exercises and other resources. A learner’s engagement may change gradually, but a simple count of activities does not reveal why the change occurred or whether it is meaningful.

02 THE RESEARCH PROBLEM

Investigate whether behavioural sequences can reveal meaningful transitions preceding sustained disengagement or changes in learning activity.

03 WHY THE PROBLEM IS NON-TRIVIAL

Reduced activity can mean many things: completion, difficulty, frustration, changing study strategy or loss of interest. Observed behaviour does not directly reveal intent, and a system that labels every reduction as disengagement would be misleading.

CENTRAL RESEARCH QUESTION

“Can behavioural sequences reveal when learning engagement begins to change in a meaningful way?”

06 THE ENVISIONED OUTCOME

A learner-behaviour analysis application that visualizes activity trajectories and surfaces candidate transitions with evidence and uncertainty.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong solution will avoid claiming to infer learner motivation from behaviour alone and will clearly define the limits of its interpretation.

04 THE RESEARCH OPPORTUNITY

Study sequential behaviour modelling, change-point detection and educational analytics. Define an operational target that can be evaluated responsibly. Compare static activity features with approaches that preserve temporal structure and examine whether detected transitions are persistent.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- What measurable event should represent an engagement transition?
- Which sequence patterns are informative?
- How can temporary breaks be separated from sustained change?
- How should individual baseline behaviour be represented?
- What evidence would justify an early-warning signal?

08 INVESTIGATION STANDARD

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Detecting Counterfeit Products Through Visual Inconsistency

PROBLEM ID	ML-T2-019	DOMAIN	Computer Vision · Product Authentication	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Counterfeit products often attempt to reproduce legitimate packaging, labels and visual identity. Differences can be extremely subtle and may involve typography, alignment, colour, texture, logos or layout.

02 THE RESEARCH PROBLEM

Investigate whether computer vision can identify visual inconsistencies associated with potentially counterfeit product imagery while accounting for legitimate variation between manufacturing batches, regions and acquisition conditions.

03 WHY THE PROBLEM IS NON-TRIVIAL

Lighting, camera angle, image quality and legitimate packaging variants can create differences that resemble counterfeit indicators. A model may also learn brand-specific backgrounds or image sources instead of product authenticity.

CENTRAL RESEARCH QUESTION	"Can visual representation learning distinguish meaningful counterfeit indicators from legitimate manufacturing variation?"
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06 THE ENVISIONED OUTCOME

A software image-analysis prototype that compares product imagery, identifies suspicious visual deviations and presents interpretable evidence for human inspection.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong solution will demonstrate robustness and will avoid presenting visual similarity as definitive proof of authenticity.

04 THE RESEARCH OPPORTUNITY

Study visual similarity, metric learning, representation learning and anomaly detection. Determine whether the selected problem is best treated as classification, verification, retrieval or another formulation. Establish a baseline and test robustness to controlled image variation.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- What should constitute visual evidence of a counterfeit?
- How can legitimate packaging variation be represented?
- Can the system compare previously unseen product instances?
- How should image acquisition differences be handled?
- Can the system show which visual regions contributed to a decision?

08 INVESTIGATION STANDARD

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Detecting Hidden Energy Waste in Buildings

PROBLEM ID	ML-T2-020	DOMAIN	Energy Intelligence · Time-Series ML	LEVEL	Advanced
DURATION	2–3 Weeks	TEAM	Assigned by Learn Depth	COST	ZERO-COST / SOFTWARE ONLY

01 THE CONTEXT

Buildings consume energy through cooling, heating, lighting, computing equipment and other systems. Excessive consumption may persist without a clear equipment failure, particularly when occupancy and environmental conditions vary.

02 THE RESEARCH PROBLEM

Investigate whether machine learning can establish expected energy behaviour and identify deviations that may indicate avoidable consumption under comparable operating conditions.

03 WHY THE PROBLEM IS NON-TRIVIAL

Energy demand changes with occupancy, weather, time of day, holidays, schedules and building characteristics. A simple threshold can flag legitimate high-demand periods, while an overly flexible model may absorb genuine inefficiency into its definition of normal.

CENTRAL RESEARCH QUESTION	“Can an ML system distinguish genuine energy inefficiency from legitimate changes in building behaviour?”
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06 THE ENVISIONED OUTCOME

A software energy-intelligence application that compares observed and expected consumption, highlights persistent deviations and provides supporting evidence for investigation.

07 WHAT WOULD COUNT AS A STRONG RESULT?

A strong solution will connect anomaly detection to a defensible definition of expected energy behaviour rather than simply flagging high consumption.

04 THE RESEARCH OPPORTUNITY

Study energy forecasting, baseline modelling, anomaly detection and contextual normalization. Determine how expected consumption should be defined and how the model can distinguish a legitimate change in conditions from unexplained excess use.

05 QUESTIONS THE INVESTIGATION SHOULD ANSWER

- What contextual variables are essential for defining expected consumption?
- How should building occupancy or schedule be represented?
- What constitutes waste rather than ordinary demand variation?
- How should seasonal changes be handled?
- Can the system estimate the magnitude and persistence of potential waste?

08 INVESTIGATION STANDARD

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